MNE python.

python modules: numpy, mne

① loading data
→ mne is used to read data from neuroscience devices
→ there are many different systems, MNE gives a unified way to load all of them.
→ sample_data_folder = mne.datasets.sample.data_path()

these are built in datasets.
contains EEG + MEG recordings
→ .fif (MNE's native file format)
→ then we build the full path to a specific data file using the / operator.
raw = mne.io.read_raw_fif(sample_data)
→ creates an object which contains the sample data in mne format

output: prints out info about the file
→ ssp projectors: used to remove extra noise through linear matrices.
→ recorded in empty room recordings to identify and remove environmental magnetic noise.
→ average of all channels = 0

PSD (power spectral density)
→ taking fourier transforms of wave signals.
→ shows which frequencies are stronger in a signal.
compute_psd → computes frequency
uses Fourier transform

excludes = "bads"
→ noise interruptions.

Preprocessing

ICA → independent component analysis
→ blind source separation technique
→ based on non-gaussianity

creating ica object
ica = mne.preprocessing.ICA(n_components=20, random_state=97, max_iter=800)

→ initialization to fix randomness

reduces 60+ channels to near 20

stabilize
(200 for less complex data)

$$\boxed{X = AS}$$

→ mathematical concept of ICA

ica.fit(raw)
→ (.fit() is used in python to train ML models)

Note: ICA and SSP are both different cleaning methods, ICA being more advanced of the two.

PSD is an analysing method which shows power vs frequency²

make a copy of the original signal to make analysis
orig_raw = raw.copy()

ica. apply (raw)

removes components listed in exclude

so, fit (raw) is used to identify whats to exclude
apply (raw) actually excludes

major usage of ICA is to remove eye blinks

Detecting Experimental Events

stim channels: store event markers not just brain signals
unlike EEG signals (smooth waves) they look like square pulses (sudden jumps)

STI 014 → standard trigger channel

events = mne . find_events (raw, stim_channel = "STI014")
an array that stores when events occur.

each array looks like:
[sample_idex, previous_value, event_id]

this is the event type

we need to define different event types
this is done by event_dictionary

Epochs

→ a small time window around an event
→ epoch objects require ⟶ raw
⟶ events

→ cutting data into small chunks around events and then using rejection
→ rejection is done by creating a dictionary which consists of threshold frequency.
→ epoch is very different from ICA

each epoch includes:
pre stimulus baseline
post stimulus brain response
→ we categorize auditory and visual data
signals will look different in shape / size

Time Frequency Analysis

mne.time_frequency. computes PSD, TFR etc

Time frequency representation: power vs time vs frequency

Morlet wavelets are small wave like signals used to detect specific frequencies in data over time

→ this will give a heatmap.

Estimating Evoked response
evoked response ⟶ average response
when we average: signal stays
noise cancels.

legend (used for plotting)

Global field power: how strong is the overall brain response across all sensors at a given time.

Inverse modelling

→ Used to locate where the brain signal is coming from.
→ MNE: minimum norm estimation
→ using MRI scans we define thousands of points on the brain surface. These are possible "sources"

MNE uses a linear inverse operator to project sensor measurements to source space. we can load a pre-computed inverse operator.

→ signal to noise ratio is used to smooth.

mne. minimum_norm. read_inverse_operator is a precomputed mapping tool.

there are other methods beside MNE like dSPM or loreta

OUTPUT FINAL:

a dynamic brain map showing which areas activate over time.